

The Science Behind CPTG

A Framework of Scales for Gravity, Structure, Cosmology, and Nuclear Reactions

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Abstract

Curvature Polarization Transport Gravity (CPTG) is a scale-structured framework that asks whether physical discrepancies appearing in bound systems, cosmology, primordial nuclear reactions, and compact gravity can be organized through one source–polarization–transport discipline rather than through separate adjustable remedies. The framework does not require one reduced equation to describe every regime. Instead, each scale uses a declared physical realization whose formulas, exponents, normalizations, and projection maps are fixed before the target calculation is performed.

On gravitational scales, General Relativity remains the foundation. CPTG adds curvature polarization and curvature transport where additional geometric organization becomes important. In bound systems, the structural acceleration scale $a_\star$ belongs to the object, while Structural Mode N is measured only after the field has been solved. In compact gravity, infinite-curvature singularities are excluded as physical states and finite continuation must recover the successful General Relativistic exterior. At cosmological scale, one native π -branch is viewed through acoustic, luminosity, and other observational projections tied to the same geometric phase. The locked CMB construction also achieves comparison-map closure in real Planck and WMAP temperature data: its fixed angular-power envelope is translated into the measured harmonic sky coordinate and reconstructed in map space without dataset-specific theory retuning.

At primordial nuclear-reaction scale, Vertex, Bridge, Closure, and Saturation organize the conserved light-nuclide state space, while a separate live mass-seven transport law acts inside the evolving network. Nuclear dynamics remain nuclear dynamics; the gravitational field equation is not inserted into nuclei. Across all of these regimes, observations supply physical inputs or test outputs, but they do not retune the governing formulas after the result is known.

1 What CPTG Is

Some of the hardest problems in modern physics appear when familiar descriptions are pushed across very different scales. Galaxies show more gravitational support than their visible matter appears to provide. Compact objects force General Relativity into an extreme regime in which the classical continuation approaches an infinite-curvature singularity. Cosmological measurements made through different observational channels do not always return the same inferred coordinates. Primordial nuclear reactions retain a persistent lithium discrepancy even where much of the underlying light-element network succeeds. These problems are usually studied separately, but they raise a common question: does each discrepancy require its own new ingredient, or has part of the physical description been left unresolved?

Curvature Polarization Transport Gravity (CPTG) develops the second possibility. General Relativity remains the gravitational foundation, while CPTG asks whether additional geometric organization becomes important where a reduced gravitational description no longer completes the physical picture. Curvature polarization and curvature transport provide that additional structure [6]. In galaxies, they can organize gravitational response across a large bound system. In compact objects, CPTG retains the successful General Relativistic exterior while requiring the physical continuation to remain finite rather than terminate at an infinite-curvature singularity.

CPTG also extends beyond the gravitational field problem. In primordial nuclear-reaction state space, it does not insert the gravitational field equation into nuclei. Instead, the same source–polarization–transport discipline is realized through a conserved reaction system whose native dynamics remain nuclear. Free nucleons, deuterium, the tritium/helium-3 closure pair, and helium-4 occupy the Vertex, Bridge, Closure, and Saturation sectors, while mass seven is treated by its own live transport action in the evolving network. The common element across scales is therefore not one force equation, but one organizing discipline adapted to the physical variables, conservation laws, and native dynamics of each regime.

The same discipline governs computation. A CPTG reduction or state-space construction is registered before evaluation: its formulas, exponents, normalizations, and projection maps are locked before the target calculation is run. Observations can supply the physical state of the system or test the output, but they do not rewrite the governing formula through post-result tuning knobs or dataset-specific rescue terms. If a locked reduction misses, the mismatch remains visible.

The contrast with other approaches is clearest on the gravitational side. The standard dark-matter description places the additional gravitating source in a non-baryonic component. The original MOND galaxy prescription keeps the visible matter but changes the low-acceleration relation through a universal acceleration scale a_0 [1]. CPTG instead keeps ordinary matter on the material side of the gravitational theory, keeps Einstein curvature as the foundation, and assigns the additional organized response to geometry itself.

2 A Framework of Scales

Different physical regimes expose different observables, but CPTG requires them to remain inside one scale-aware source–polarization–transport discipline. A rotating galaxy reveals gravitational support through orbital motion, a pressure-supported system through stellar dispersion, cosmological redshift through the phase of large-scale geometry, a reaction network through movement among nuclear states, and a trapped strong-field region through the requirement of finite continuation. The characteristic size, symmetry, gradient, timescale, and observable may change without requiring a separate organizing theory for each problem.

Regime	What changes physically	How CPTG organizes it
Bound gravitating systems	Size, mass distribution, rotation, pressure support, structural state	Object-dependent structural acceleration a_* ; solved polarization–transport field; post-solution Structural Mode N where defined
Extended and directional systems (for example, extended bound systems or directional mergers)	Coherence, anisotropy, collisions, directional redistribution	The same geometric response with transport retained explicitly when symmetry no longer hides direction
Compact strong fields	Curvature gradient and trapped geometry become extreme	General Relativity remains the successful exterior description; CPTG requires a finite physical continuation when additional response becomes important
Cosmology	The observable channel and the rate history used to describe one large-scale state	One native π -branch with acoustic, luminosity, and other downstream observational projections
Primordial nuclear state space	Reaction topology, conservation, and movement among bound states	A conserved polarization–transport realization built from Vertex, Bridge, Closure, and Saturation; nuclear dynamics remain nuclear dynamics

Table 1: CPTG is scale-structured rather than scale-fragmented. Gravitational regimes are reductions or projections of the gravity-side framework; the nuclear sector is a conserved state-space realization of the same organizing principle rather than a gravitational force law acting inside nuclei.

The value of this structure is not that every problem receives the same formula, but that the same organizing logic must survive while the physical realization changes. That is a stronger requirement than constructing a separate correction for each anomaly.

3 Polarization and Transport

Curvature polarization is the idea that geometry can develop an organized response around a physical source. The word *polarization* is an analogy to induced organization: it does not mean electromagnetic polarization, photon polarization, or a hidden material medium. In CPTG, it is the geometry that responds.

Curvature transport describes how that organized response is carried or redistributed. In a highly symmetric system, much of the transport structure can be compressed into a simple radial description. In a directional or changing system, the transport itself can become one of the important observables. Polarization answers, “how strongly is the

geometry responding here?” Transport answers, “how is that response organized and carried through the system?”

Together they provide a different place for gravitational structure to live. Instead of saying that every excess field must correspond to extra unseen mass, CPTG asks whether the field can contain organized geometric response that is sourced by the ordinary physical system.

General Relativity and CPTG are therefore complementary inside the framework rather than competing replacements. The Einstein geometry supplies the foundation everywhere. Where polarization–transport structure is negligible, the GR description is sufficient. Where that structure becomes important, CPTG resolves the additional geometry instead of interpreting the discrepancy as evidence that Einstein’s law itself has stopped working [6].

4 Structural Scale $a_\star$ and Mode N

4.1 The structural acceleration scale $a_\star$

One of the clearest differences between CPTG and the original MOND galaxy prescription is the meaning of the acceleration scale.

The original MOND prescription is famous for asking whether one universal low-acceleration scale can organize galaxy dynamics. CPTG instead treats $a_\star$ as a property of the gravitating structure. It is an object-dependent structural acceleration scale, not a universal constant that every system is required to share [6].

The distinction matters because galaxies with different mass distributions, sizes, angular momenta, pressure support, or evolutionary states need not organize around exactly the same acceleration number. CPTG allows the physical state to differ while keeping the governing law fixed.

Object dependence does not make $a_\star$ a tuning license. In the registered reductions, the object supplies its measured structure and kinematics, the same CPTG response law is applied, and the structural scale is inferred as part of that physical solution. The law is common even when the physical state is not.

That distinction is central to a framework of scales: a universal constant assigns one acceleration scale to every system, whereas $a_\star$ allows each gravitating object to carry the structural acceleration scale appropriate to its physical state.

4.2 Structural Mode N and CSMI

Structural Mode N answers a different question from a_* . It is not used to determine the gravitational fit. It is read *after* the CPTG field has been solved and measures where the important structural change in that solved field is concentrated [7].

Its defining structural ratio is

$$N = \frac{R}{\lambda}, \quad (1)$$

where R is the declared outer radius of the evaluated solved-field domain and λ is the weighted structural radius extracted within that domain. The ratio is dimensionless: larger N places the weighted structural organization farther inward relative to R .

That makes N a theory-generated structural coordinate. Conventional galaxy classification usually begins with appearance: an observer looks at the luminous system and assigns a descriptive family. CPTG reverses the direction. The gravitational field is solved first, without feeding a Hubble type or morphology label into the mode calculation. Only then is the field’s internal organization measured.

The Curvature Structure Mode Index, or CSMI, gives that continuous mode a practical structural language. The CSMI value is the solved N itself; the associated CSMI type maps ranges of N to descriptive structural families. Its purpose is not to improve the rotation-curve solution after the fact. Its purpose is to let CPTG generate a structural catalog from the geometry it has already solved and then compare that catalog with the visible organization of real galaxies [7].

In public-facing terms, N asks, “where is the solved curvature structure organized?” CSMI asks, “what recognizable structural family does that organization resemble?” The classification comes after the physics rather than being supplied to it.

A rotation curve tests whether the field amplitude is plausible, while Structural Mode and CSMI test whether the same solved field contains coherent organization that was not inserted as a classification label beforehand. The existing structural validation shows strong population ordering without supplying conventional morphology to the field solution [7]. The quantities should therefore be read together but not conflated: a_* describes the structural acceleration state, N describes the organization of the solved response, and CSMI turns that organization into a readable structural catalog.

5 The Cosmological Contrast

The standard Λ CDM framework remains extraordinarily successful as a global cosmological description, yet several precision and cross-scale pressure points remain unresolved. Galaxy dynamics require large inferred non-baryonic components; early- and late-universe routes to the present expansion scale do not agree as closely as expected [3, 4]; very mature high-redshift structures continue to test formation timelines; primordial lithium remains anomalous while deuterium and helium are much more successful [5]; and the horizon

problem requires an explanation for the remarkable uniformity of the cosmic microwave background. These questions are usually assigned different pieces of physics because they arise in different regimes.

MOND-style galaxy dynamics posed one of the clearest empirical challenges to the dark-matter interpretation by showing that a simple low-acceleration rule captures striking regularities in galaxies. For the comparison made here, the original MOND prescription remains primarily a galaxy-dynamics benchmark rather than a complete replacement for the standard cosmological, nuclear, and strong-field frameworks. CPTG takes the empirical lesson seriously—galaxies contain deep acceleration regularities—but places that lesson inside a broader geometric architecture rather than stopping at one universal acceleration threshold.

The relevant contrast is therefore not between a simple theory and a complicated one, but between different choices about where physical freedom is placed.

Question	Λ CDM / dark-matter	MOND-style dynamics	CPTG
Where does the additional galaxy-scale gravity live?	Primarily in an inferred non-baryonic matter distribution	In a modified low-acceleration relation	In additional geometric polarization–transport response sourced by the physical system
Acceleration scale	No CPTG-like object structural scale is required by the standard gravitational law	A universal a_0 is central	$a_\star$ belongs to the object and is inferred under a common law
How can different objects vary?	Halo mass, concentration, profile, baryonic history, and other model ingredients can vary	The universal acceleration law is retained while baryonic structure varies	The material source and structural state vary; the geometric response law is retained
Cosmology	One standard cosmological model with dark matter and Λ or dark energy	No single MOND cosmology has displaced Λ CDM as the standard model	One native geometric history with multiple observation-facing projections
Additional structural output	Halo parameters and derived environmental relations	Primarily the modified dynamical response	Post-solution Structural Mode N , CSMI structural classification, and other geometric readouts

Table 2: The main difference is not whether a framework has parameters. It is where the unexplained response is placed, what is universal, what is allowed to be object-dependent, and whether the same architecture continues beyond galaxy rotation curves.

Standard dark-halo modeling represents real system-to-system diversity through halo mass, concentration, profile, baryonic history, and related quantities. That flexibility is often physically motivated, but it places explanatory freedom in both visible and unseen model components. CPTG places that freedom differently. It has no dark halo in the material source that can be reshaped until the local field agrees. The geometry has to do the work.

The galaxy benchmark shows why this is scientifically interesting. In the registered primary SPARC comparison of 153 galaxies and 3168 resolved rotation-curve points, CPTG lowers total χ^2 by 96.016% relative to the fixed- a_0 MOND benchmark, lowers the velocity RMS residual by 67.86%, lowers the mean absolute velocity error by 74.78%, and gives the lower galaxy-level χ^2 in 146 of 153 systems (95.42%). These percentages describe differences

in fit statistics under the declared SPARC comparison; they are not probabilities that either theory is correct. As a separate fairness audit, the MOND benchmark is then granted stellar mass-to-light freedom, allowing its stellar contribution to adjust rather than remaining fixed at the baseline value. The CPTG governing response law is not changed for that audit, and the CPTG advantage remains rather than disappearing under the more flexible MOND comparison [2, 7, 19].

Those results do not make galaxy rotation the whole theory. They demonstrate something more useful: the scale-dependent structural idea is already capable of outperforming the fixed- a_0 MOND benchmark in the regime where that benchmark is strongest.

6 One Native Branch, Acoustic and Luminosity Projections

At cosmological scale, CPTG separates the underlying Universe from the different ways we observe it. There is only one native cosmological history: the native π -branch. The acoustic and luminosity descriptions are projections of that branch, not separate universes and not competing versions of reality. They are different observational views of the same native geometric state.

A simple analogy is lightning. A single lightning discharge is the event itself. An observer may see the flash and later hear the thunder. The flash and the thunder come from the same event, but they reach the observer through different physical channels and therefore carry different timing information.

CPTG uses that picture in the following way:

- **Native π -branch:** the underlying geometric history of the Universe. This is the physical state from which the observational projections are derived.
- **Acoustic projection:** the way that same native geometry appears through acoustic observables. It carries the sound-horizon and CMB-facing response of the native state.
- **Luminosity projection:** the way that same native geometry appears through light-based and luminosity-distance observations. It carries the distance and brightness response of the same native state.

In the lightning analogy, the native π -branch is the electrical discharge, the luminosity projection is the flash, and the acoustic projection is the thunder. The analogy is not meant to say that cosmological projection timing is literally caused by the travel-time difference between light and sound. Its purpose is simpler: one physical event can be observed through different channels, and those channels do not have to return identical timing or rate information.

The framework therefore contains one native cosmological π -branch as the underlying geometric history. The acoustic and luminosity projections are derived from that native branch rather than being independently initialized cosmologies. Growth, lensing, and

other comparison quantities are likewise downstream readouts of the same native history [8, 6].

A common geometric phase identifies the same stage of the native history. The acoustic and luminosity projections are then evaluated at that same phase using their own rate histories. This means that two observations can refer to the same underlying cosmic state while returning different accumulated times or different expansion-rate coordinates.

The CMB provides an important example of the acoustic side of this structure. The CMB comparison construction associated with the locked native π -branch has been translated into real Planck and WMAP temperature-map coordinates. The observed CMB supplies the observational field coordinate, while the locked CPTG construction supplies the angular-power structure carried into that coordinate. The resulting comparison-map closure shows that the native CPTG geometry can be expressed successfully through the CMB-facing acoustic projection without changing the underlying native branch [14].

This distinction changes how apparent cosmological disagreements are interpreted. If an acoustic measurement and a luminosity-distance measurement return different present-day expansion coordinates, CPTG does not immediately treat them as two failed measurements of one number. It first asks whether they are two different observational projections of the same native geometric state.

In short:

$$\text{one native geometry} \longrightarrow \begin{cases} \text{acoustic projection,} \\ \text{luminosity projection.} \end{cases} \quad (2)$$

The native branch tells us *what the Universe is doing geometrically*. The acoustic and luminosity projections tell us *how that same geometric state is measured through different observational channels*.

7 Time Across the Native Branch and Its Projections

Time is part of the measurement definition in CPTG because a common native geometric phase does not require every observational projection to accumulate the same elapsed time.

The native π -branch defines the geometric phase sequence. The acoustic and luminosity projections are evaluated along that same sequence, but they do not have to accumulate the same time while moving through it. The native branch and each observational projection have their own rate histories and therefore their own clocks [8, 9, 10]. The timing difference is part of the measurement information rather than a discrepancy to be erased.

That is especially important in cosmology because quantities such as an expansion rate, a distance, or an inferred age are not measured in isolation from the channel that produced

them. In CPTG, a redshift first selects the common native geometric phase. The acoustic or luminosity description is then evaluated at that same phase using the rate history of that projection. Two measurements can therefore refer to the same underlying cosmic state while carrying different accumulated times and different projection-specific rate coordinates.

This is why CPTG does not simply borrow a conventional cosmological age and call it native time. A Λ CDM age can be used as an external comparison coordinate, but it does not overwrite the native or projection clocks used by CPTG. The same caution applies in the opposite direction: an acoustic clock cannot automatically be substituted for a luminosity clock simply because both describe the same redshift.

High-redshift galaxies provide a direct cross-scale test because their cosmological placement and their local gravitational structure come from different observables. In CPTG studies of the compact quiescent galaxy NMBS-C7447 at $z = 1.80$ and the pressure-supported slow rotator XMM-VID1-2075, redshift fixes the cosmological phase while stellar dispersion, size, and mass separately constrain the bound-system state. These cases test whether a mature galaxy can be evaluated at an earlier native phase without changing the local structural law to accommodate the redshift [15, 16].

The native/projection timing distinction is not faster-than-light travel and it is not time travel. It is a statement about the relationship between physical state and measurement. In CPTG, equal phase means “the same place in the underlying geometric history.” Equal phase does not require equal accumulated clock time in every observational projection. That timing structure is one of the reasons the Hubble tension can be treated as a bridge between measurements rather than only as a disagreement to be averaged away.

8 The Hubble-Tension Bridge

The Hubble tension provides a direct example of the CPTG projection structure because the lower expansion scale inferred through the early-universe acoustic channel and the higher scale inferred through the late luminosity-distance channel are usually treated as a disagreement that must be removed, shifted, or explained within one cosmological parameter set [3, 4]. CPTG instead asks whether both measurements can be valid observational projections of one native geometry.

The native π -branch lies between the two familiar observational endpoints. Its derived acoustic projection falls very close to the lower CMB-facing expansion scale. The luminosity side is represented by a late distance-ladder comparison coordinate on the same native phase history. In other words, CPTG does not solve the tension by averaging the two observations into a compromise number. It builds a geometric bridge between the channels [9].

Under this interpretation, what appears inconsistent in a one-clock, one-coordinate description becomes a measurable separation between projections of the same native geometry.

The acoustic side of the bridge is derived from the native CPTG structure. The present luminosity normalization is still conditioned by the local distance-ladder measurement, so a first-principles luminosity projector remains an important cosmology target. But the architectural result is already clear: CPTG can preserve both observational channels without pretending that they are identical measurements made through identical physics [9, 6].

9 The Horizon Problem

The cosmological horizon problem is usually explained as a communication puzzle: distant regions of the cosmic microwave background appear remarkably uniform even though a simple single-clock picture does not give them enough early time to exchange light and equilibrate. Inflation is the standard early-Universe mechanism used to address this causal problem [3]; CPTG investigates a different geometric route.

CPTG reorganizes the question. The acoustic background is not treated as a collection of independently initialized acoustic regions that later have to discover one another. It is a projection of one native geometric background. The horizon construction then asks whether the region that becomes the observed last-scattering sphere lies inside one native curvature-transport domain at the relevant phase. In the CPTG background calculation, the containment and projection conditions are simultaneously satisfied [10].

Under this construction, uniformity does not have to be created by forcing many independent acoustic patches to synchronize under one universal clock. The common acoustic background is inherited from one native geometric state, replacing a late synchronization story with a common-origin description of the background.

The same native/projection timing logic changes how age arguments are interpreted at high redshift. Redshift selects native phase first, and the time assigned to that phase depends on the clock associated with the native or observational coordinate being used. This does not replace the astrophysics of star formation, feedback, or quenching; it changes the cosmological time coordinate against which those processes are compared.

10 CMB Map Closure: From Geometry to the Measured Sky

One of the clearest real-world tests of the cosmological construction comes from the cosmic microwave background. A CMB temperature map is not merely a colored picture of hot and cold regions. It is the real-space representation of angular structure distributed across spherical-harmonic multipoles and modes. The CPTG comparison-map test asks whether a geometric construction fixed before map evaluation can be carried into that measured field without rewriting the theory to follow the data.

The observed CMB supplies the harmonic sky coordinate, while the locked CPTG construction supplies the angular-power envelope carried through that coordinate. In plain terms, the measured sky fixes where the observed hot and cold structures occur; CPTG

supplies how much angular power belongs at each retained scale. The test therefore asks whether that fixed CPTG scale-by-scale structure can inhabit the real CMB field successfully, not whether CPTG independently predicts the observed harmonic phases. The translated field is then reconstructed as a temperature map. Planck is treated through the same comparison rule, so the result is not judged by visual resemblance alone. In the primary SMICA comparison, the reconstructed CPTG field reaches an observed-field correlation of about 0.9834 with fitted residual RMS about $18.22\,\mu\text{K}$. Across the five principal Planck component products, the mean CPTG–Planck residual-RMS separation is about $0.0235\,\mu\text{K}$ [14].

The closure is also stable when the observational realization is changed. Planck full, half-mission, and odd/even split products give a mean full-band separation of about $0.0239\,\mu\text{K}$ with a total split span of about $0.00069\,\mu\text{K}$. WMAP low- ℓ bands, smoothing controls, masks, and sky-fraction changes preserve the same broad result. This matters because the comparison is not tied to one rendering of one Planck map.

The strongest safeguard is the null-envelope test. Flat, power-law, and shuffled angular-power envelopes are translated through the same observed harmonic sky scaffold and the same fitting rule. They do not reproduce the CPTG–Planck closure: shuffled spectra give residual RMS near $95.5\,\mu\text{K}$, compared with about $18.23\,\mu\text{K}$ for the Planck benchmark and $18.25\,\mu\text{K}$ for the locked CPTG envelope [14]. Sharing the observed harmonic scaffold therefore does not guarantee success for the tested generic envelopes. The angular organization carried by the fixed geometry remains essential.

Where standard CMB numerical-solver coordinates are used in the detailed comparison construction, they serve as translation coordinates for the observational envelope; they are not adopted as CPTG’s native matter content or governing cosmology.

The significance is consequently deeper than producing a Planck-like heat map. A fixed CPTG geometric construction survives translation into one of the richest measured fields in cosmology, across hundreds of retained angular multipoles, and returns a reconstructed field that remains closely adjacent to the Planck benchmark without dataset-specific theory retuning. The comparison map is the visible endpoint of that field-coordinate closure.

11 Vertex, Bridge, Closure, Saturation

At primordial nuclear-reaction scale, CPTG uses polarization and transport to organize a conserved reaction state space without treating nuclear reactions as gravitational or applying the galaxy equation to nuclei [11].

The validated light-sector architecture has four structural sectors:

Sector	Physical content	Structural role
Vertex	Free neutron-proton state	The unbound starting sector from which the light-reaction network develops
Bridge	Deuterium	The first stable bound bridge connecting free nucleons to the mass-three sector
Closure	Tritium and helium-3	A mass-three sector with two internal charge orientations retained through polarization
Saturation	Helium-4	The maximally closed terminal reservoir within the validated four-sector coordinate

Table 3: The four-sector light-nuclide architecture. Ordered transport runs through Vertex, Bridge, Closure, and Saturation, while an independent polarization direction preserves internal charge information required by exact baryon and electric-charge conservation.

The names matter less than the organization they encode. Reaction flow has a topology, conservation restricts what can move where, transport follows the ordered chain, and polarization retains information that would otherwise be lost when several microscopic species are grouped into one structural sector. The resulting source space contains three transport directions and one independent polarization direction [11].

The nuclear realization demonstrates the scale-framework principle without claiming that gravity acts inside the nucleus as it does in a galaxy: source, polarization, transport, conservation, and projection can recur as organizing structures inside a completely different native dynamical system.

12 The Lithium Transport Solution

The cosmological lithium problem isolates a selective discrepancy within otherwise successful primordial nucleosynthesis: deuterium and helium are reproduced well, while too much mass seven survives relative to the low lithium abundance inferred from old metal-poor stellar populations.

CPTG treats that selectivity as a clue. Rather than changing the whole baryon density, inserting a freely fitted lithium depletion factor, or disturbing the successful deuterium and helium channels, the CPTG lithium construction treats surviving mass seven as a transport problem inside the live reaction network [12].

The mass-seven transport rule is locked before network evolution. It acts on the live lithium-7 and beryllium-7 channels with a geometric optical depth fixed by $\ln \pi$, rather than being adjusted until the endpoint abundance looks right. In the native PRIMAT reaction-network validation, the mass-seven abundance is driven into the observational comparison band while deuterium, helium, helium-3, and the transported radiation coordinate remain essentially unchanged; all pass/fail criteria fixed before the calculation are satisfied [12].

The result matters beyond lithium because it illustrates the CPTG scale discipline: a persistent discrepancy is not automatically met with another free abundance parameter. Instead, a specific transport structure derived and frozen upstream must correct the problematic channel without damaging the channels that were already working.

The four-sector nuclear theory then generalizes that idea beyond one abundance problem. Vertex, Bridge, Closure, and Saturation provide a reusable structural language for the light reaction network, while native nuclear codes supply the actual reaction dynamics [11].

13 Finite Strong-Field Continuation

At the compact end of the scale hierarchy, CPTG retains the successful General Relativistic exterior and does not treat a black-hole horizon as a switch from one theory to another. CPTG rejects every infinite-curvature singularity as a physical state; the strong-field question is therefore how the geometry continues finitely through the region in which a classical GR-only continuation would otherwise become singular.

CPTG’s present compact-sector position is an admissibility principle: every physical compact solution must remain finite, contain no infinite singularity, conserve the required geometric structure, and match smoothly back to the appropriate General Relativistic exterior where the additional response becomes negligible [13].

Vanishing stars give that compact requirement a direct observational target. NGC 6946-BH1 and M31-2014-DS1 are massive-star disappearance candidates whose weak or absent normal explosions, optical disappearance, red or dusty remnants, and quiet high-energy signatures have been examined as possible finite structural-collapse endpoints in CPTG. Those observations do not establish CPTG causation or reveal the hidden interior; they provide real compact-endpoint records against which the finite-continuation requirement can be tested [17, 18].

The same logic appears at galaxy and compact scales in different forms: galaxy-scale geometry can carry additional organized response instead of requiring invisible matter, while compact geometry must remain finite as the field becomes extreme. The detailed interior solution remains a strong-field derivation problem, but the admissibility requirement is already fixed.

14 One Framework, Many Pressure Points

The broadest claim of CPTG is that problems appearing at very different scales can be approached through one organizing discipline without requiring every open problem to be already finished. The value of that claim is tested by applying the framework to physically different regimes and asking whether each one can be handled with its own appropriate variables while preserving the same source–polarization–transport logic.

At galaxy scale, the immediate test is the missing-gravity problem. CPTG keeps the observed baryonic matter as the material source and places the additional response in organized geometry. The registered SPARC comparison tests that response across resolved rotation curves, while the post-solution Structural Mode N and CSMI ask whether the solved field contains coherent internal organization that was not supplied as an input label

[7, 19]. Pressure-supported and high-redshift systems extend the same bound-system logic beyond ordinary nearby rotating disks: in NMBS-C7447 and XMM-VID1-2075, redshift fixes cosmological placement while stellar dispersion, size, and mass separately constrain the local gravitational state [15, 16].

At cosmological scale, CPTG separates one native geometric history from the observational projections through which that history is measured. The Hubble-tension construction compares acoustic and luminosity coordinates tied to one native phase, while the horizon construction treats the observed acoustic background as a projection of one native geometric origin [9, 10]. The CMB comparison-map program adds a direct observational field-translation test: the locked CPTG CMB envelope is carried into the measured Planck and WMAP harmonic sky coordinate, reconstructed in map space, and compared there without changing the underlying native branch [14].

At primordial nuclear-reaction scale, the framework changes mathematical form without reusing the gravitational field equation. Vertex, Bridge, Closure, and Saturation organize the conserved light-element state space, and the independently fixed mass-seven transport law acts inside the live reaction network [11, 12]. The lithium calculation is therefore a direct test of whether a locked transport structure can act on the discrepant channel while leaving the already successful deuterium and helium channels intact.

Compact gravity tests the opposite end of the scale hierarchy. CPTG retains the successful General Relativistic exterior but rejects every infinite-curvature singularity as a physical endpoint. Vanishing-star candidates such as NGC 6946-BH1 and M31-2014-DS1 provide observational records against which a finite structural-collapse endpoint can be compared without claiming that the electromagnetic record uniquely reveals the hidden interior [13, 17, 18].

These applications are deliberately different. Their significance lies in whether one scale-structured framework can move among them without introducing a separate non-baryonic dark-matter sector or adding a new adjustable remedy whenever a result becomes inconvenient.

15 A Scalable, Locked Framework

What makes these applications one framework rather than a collection of unrelated models is the order in which CPTG is used. Each calculation begins by identifying the physical regime, the relevant source, the observables, the required conservation laws, and the scale-appropriate CPTG realization. Only after those choices are declared is the calculation evaluated.

The governing formulas are then locked. Exponents, normalizations, projection maps, transport rules, and other theory-side quantities are fixed before the target result is known. Observations may determine the physical state of an object or provide the data against which an output is tested, but they do not supply post-result coefficients that rewrite the law. If a registered reduction misses the data, the mismatch remains part of the evidence rather than becoming a reason to retune the formula.

Object dependence is therefore not the same as theory dependence. A galaxy can have its own structural acceleration a_* because galaxies occupy different physical states, while the response law used to infer that state remains common. Structural Mode N is more constrained still: it is measured only after the field has been solved and cannot be used as an input classification to manufacture the solution. The same distinction appears elsewhere in the framework, where projection-specific observational coordinates, nuclear state variables, and compact-state conditions belong to the physical realization rather than to an adjustable rescue layer.

This locked-computation discipline also distinguishes CPTG from parameter-optimized inference structures. Standard Λ CDM analyses estimate cosmological and nuisance parameters from data, and galaxy-scale dark-matter modeling commonly includes halo and baryonic degrees of freedom. CPTG uses a different inference structure: observational quantities may be measured or inferred, but the registered theory-side formulas are fixed before the target comparison and are not re-optimized within each data sector. Its scalable formulas must survive the computation as written.

16 Conclusion

CPTG treats gravitational, cosmological, nuclear-reaction, and strong-field discrepancies as scale-dependent problems that can be approached through one organizing discipline without forcing every regime into the same reduced equation. On the gravitational side, ordinary matter remains the material source, General Relativity remains the foundation, and curvature polarization and curvature transport provide additional organized geometric response where the reduced Einstein description is incomplete.

The same scale discipline extends beyond galaxy rotation. Bound systems admit object-dependent structural states and post-solution diagnostics; cosmology is organized by one native π -branch and its observational projections; the CMB map-closure test translates the locked geometric envelope into the measured harmonic sky coordinate and reconstructs it in real map space; primordial nuclear reactions realize the framework in conserved state space; and compact gravity requires finite continuation rather than an infinite-curvature physical endpoint.

Across these scale-specific realizations, the governing formulas are fixed before computation. Observations determine physical inputs or test outputs, not post-result theory adjustments. The central CPTG claim is therefore not that every unresolved problem has already been closed, but that one scale-structured geometric discipline can be tested from one regime to the next without changing its rules after the answer is known.

Data, Code, and Evidence Availability

This article is a public-facing synthesis of the CPTG theory architecture. Detailed derivations, registered benchmark protocols, numerical evidence, source assumptions, native-branch and projection constructions, reaction-network qualifications, and failure conditions are retained in the cited CPTG research papers. Supporting models, benchmark implementations, manuscripts, and research materials are maintained in the public CPTG repository: <https://github.com/CLG2025/CPTG>.

References

- [1] M. Milgrom, *A Modification of the Newtonian Dynamics as a Possible Alternative to the Hidden Mass Hypothesis*, *Astrophys. J.* **270**, 365–370 (1983), DOI: <https://doi.org/10.1086/161130>.
- [2] F. Lelli, S. S. McGaugh, and J. M. Schombert, *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves*, *Astron. J.* **152**, 157 (2016), DOI: <https://doi.org/10.3847/0004-6256/152/6/157>.
- [3] Planck Collaboration, *Planck 2018 Results. VI. Cosmological Parameters*, *Astron. Astrophys.* **641**, A6 (2020), DOI: <https://doi.org/10.1051/0004-6361/201833910>.
- [4] A. G. Riess et al., *A Comprehensive Measurement of the Local Value of the Hubble Constant with $1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ Uncertainty from the Hubble Space Telescope and the SH0ES Team*, *Astrophys. J. Lett.* **934**, L7 (2022), DOI: <https://doi.org/10.3847/2041-8213/ac5c5b>.
- [5] B. D. Fields, *The Primordial Lithium Problem*, *Annu. Rev. Nucl. Part. Sci.* **61**, 47–68 (2011), DOI: <https://doi.org/10.1146/annurev-nucl-102010-130445>.
- [6] Carter L. Glass Jr., *Curvature Polarization Transport Gravity: A Unified Geometric Framework for Structure, Cosmology, and Nuclear Reactions*, 2026, DOI: 10.5281/zenodo.22441039.
- [7] Carter L. Glass Jr., *Structural Mode N in CPTG: Galaxy Structural Classification from the Solved CPTG Field*, 2026, DOI: 10.5281/zenodo.22436682.
- [8] Carter L. Glass Jr., *CPTG Geometric Pi Branch: Cosmological Comparison Coordinates*, 2026, DOI: 10.5281/zenodo.22426857.
- [9] Carter L. Glass Jr., *CPTG and the Hubble-Tension Bridge*, 2026, DOI: 10.5281/zenodo.22429502.
- [10] Carter L. Glass Jr., *CPTG and the Cosmological Horizon Problem: Native Origin, Geometric Containment, and Acoustic Projection*, 2026, DOI: 10.5281/zenodo.22430125.
- [11] Carter L. Glass Jr., *A Universal Geometric Theory of Nuclear Reactions in CPTG: The Four-Sector Foundation of Vertex, Bridge, Closure, and Saturation*, 2026, DOI: 10.5281/zenodo.22439971.
- [12] Carter L. Glass Jr., *The Cosmological Lithium Problem and the CPTG Lithium Solution*, 2026, DOI: 10.5281/zenodo.22420032.
- [13] Carter L. Glass Jr., *Curvature-Bounded Black Holes and Compact Objects in CPTG: Strong-Field Admissibility, Polarization–Transport Crossover, and Research Roadmap*, 2026, DOI: 10.5281/zenodo.22424445.
- [14] Carter L. Glass Jr., *CMB Comparison-Map Closure in CPTG: Real Planck and WMAP Temperature-Map Tests*, 2026, DOI: 10.5281/zenodo.22413491.
- [15] Carter L. Glass Jr., *NMBS-C7447 at $z=1.80$ in CPTG: Native-Phase Placement, Cross-Scale Locking, and Structural Mode N*, 2026, DOI: 10.5281/zenodo.22438279.
- [16] Carter L. Glass Jr., *CPTG Native-Phase Jeans and Structural Validation: XMM-VID1-2075*, 2026, DOI: 10.5281/zenodo.22445804.

- [17] Carter L. Glass Jr., *NGC 6946-BH1 and the Vanishing-Star Problem in CPTG: A Finite Structural-Collapse Interpretation*, 2026, DOI: 10.5281/zenodo.22433701.
- [18] Carter L. Glass Jr., *M31-2014-DS1 and the Vanishing-Star Problem in CPTG: A Finite Structural-Collapse Interpretation of a Contested Stellar Endpoint*, 2026, DOI: 10.5281/zenodo.22431737.
- [19] CPTG, Supporting Python Models, Benchmark Implementations, and Research References for Curvature Polarization Transport Gravity, companion resource for the present work, available at <https://github.com/CLG2025/CPTG>.